Random Projection for Indirect Quantization

Yuefeng Cao, Wenyi Zhang
Department of Electronic Engineering and Information Science
University of Science and Technology of China
Hefei, China
yf980515@mail.ustc.edu.cn and wenyizha@ustc.edu.cn

***Abstract*—In indirect quantization tasks, the quantize-then-estimate (QE) scheme is commonly adopted when encoders are distributed or task-agnostic. In some scenarios, however, the observation dimension substantially exceeds that of the underlying state, rendering direct quantization of the observations highly inefficient. In this work, we propose a random-projection-then-quantization (RPQ) scheme. Specifically, high-dimensional observations are first projected onto a lower-dimensional representation via random projection, after which quantization is applied only to the reduced dimensions rather than to all original dimensions. This strategy reduces the required bit rate and thereby alleviates communication overhead. The rate–distortion performance of the proposed scheme is analyzed. Under low-rate constraints, the proposed scheme outperforms the conventional QE scheme and, in certain scenarios, continues to do so even at higher rates.**

***Index Terms*—Composite source, indirect quantization problem, random projection, universal quantization, rate-distortion theory.**

I. Introduction

The problem of semantic communication has long attracted attention [1], [2]. Recent studies have incorporated semantic information into the design of communication systems [3], [4]. These works are typically task-oriented, rather than aimed at directly compressing or transmitting signals, with objectives closely related to semantic information. A common model for studying semantic communication is the composite source model [5]–[7], which consists of an intrinsic state and an extrinsic observation. The intrinsic state represents the semantic information, whereas the encoder has access only to the observation. Since many practical problems involve estimating continuous latent parameters (e.g., channel coefficients, localization coordinates, phase offsets, or other physical quantities), modeling the semantic state S as a continuous random vector is both natural and necessary in such settings. In this work, the composite source model with a continuous-valued state is adopted for further investigation.

Our problem falls into the category of indirect source coding [8]–[10]. In [7], [8], the optimality of the estimate-then-compress (EC) scheme was established. In the context of quantization, the corresponding approach is the estimate-then-quantize (EQ) scheme [11], in which the state is first estimated from the observation and then quantized.

Nevertheless, in many practical applications, encoders need be universal and task-agnostic, i.e., unaware of the semantic relation between the observation and the state, making the EQ scheme infeasible. A natural alternative is the quantize-then-estimate (QE) scheme. In [12], [13], the rate–distortion risk of the compress-then-estimate (CE) scheme was analyzed, and a method for designing estimators independent of specific compression techniques was proposed.

In some cases, although the exact relationship between the state and the observation is unknown, the observation is known to have a much higher dimension than the state. In such highly overdetermined systems, directly quantizing the observation wastes rate. In this paper, we propose a random-projection–then–quantization (RPQ) scheme: the high-dimensional observation is first projected into a lower-dimensional representation and then quantized. A task-specific estimator then reconstructs the state from the quantized low-dimensional representation.

As related work, random projection has been used in signal processing for practical tasks such as computer vision [14], [15] and speech recognition [16]. These works empirically demonstrate its potential, e.g., reducing transmitted data or improving computational efficiency while maintaining classification accuracy, but do not investigate it from a rate–distortion perspective. Random projection has also been used in compressed sensing [17], which aims to recover high-dimensional sparse signals. Some works in the machine learning context use randomly projected and quantized vectors to estimate inner products [18] or provide labels for supervised learning [19]. None of these works investigate quantization with random projections in the context of indirect source coding from a rate–distortion perspective. In this work, we explicitly adopt the composite source model to formulate and analyze this problem. Different from the prior works, our formulation imposes no sparsity assumptions on the signals and treats the encoder as task-agnostic.

Our main contributions are as follows:

- We propose the RPQ scheme and characterize its rate–distortion performance for a given projection matrix in the high-rate regime.
- We establish the relationship between the expected estimation distortion and the dimension K of the low-dimensional representation vector for a general linear Gaussian composite source with a scalar state and isotropic noise, and analyze the trade-off between estimation distortion and distortion induced by quantization.
- Through numerical results, we compare the rate–distortion performance of the RPQ scheme for

different values of K with those of the QE and EQ. The results show that random projection improves rate–distortion performance at low rates and, in some scenarios, even at higher rates.

The remaining part of the paper is organized as follows:

In Section II, we describe the source model adopted in this work and formulate the problem under consideration. Section III presents the RPQ scheme and analyzes its performance, including the rate–distortion performance in the high-rate regime for a given projection matrix Φ. In Section IV, we establish the relationship between the expected estimation distortion and the dimension K of the low-dimensional representation in a commonly studied setting, and analyze the trade-off between estimation distortion and distortion induced by quantization. Section V compares the rate–distortion performance of the RPQ scheme under different values of K with those of the EQ and QE schemes via numerical results. Finally, Section VI concludes the paper.

II. Problem Formulation

A composite source consists of an intrinsic state S and an extrinsic observation X, where each state value corresponds to a distinct sub-source and distribution of X. In encoding, the encoder observes only X [7], [20], and the state can only be inferred from it. In this paper, we focus exclusively on indirect quantization tasks, requiring only the reconstruction of S.

In this paper, we consider composite sources where $S \in \mathbb{R}^m$ and $X \in \mathbb{R}^n$ are zero-mean and jointly Gaussian, related by $X = HS + W$, with H an $n \times m$ matrix and $W \sim \mathcal{N}(\mathbf{0}, \Sigma_W)$ independent of S. Denote the joint covariance matrix of (S, X) as

$$\begin{bmatrix} \Sigma_S & \Sigma_{SX} \\ \Sigma_{SX}^T & \Sigma_X \end{bmatrix}. \tag{1}$$

We have $\Sigma_{SX} = \Sigma_S H^T$ and $\Sigma_X = H\Sigma_S H^T + \Sigma_W$.

The optimality of the EC scheme under the mean-squared error (MSE) criterion was proved in [7], [8]. In the EC scheme, the state S is first estimated as $\tilde{S}$ from the observation X, which is then compressed into $\hat{S}$ and delivered to the destination. The procedure is illustrated in Figure 1.

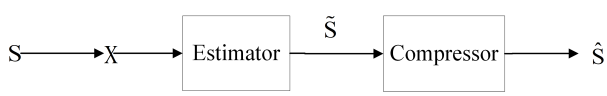


Fig. 1. The procedure of the EC scheme

When the encoder is task-agnostic, the EC scheme cannot be implemented. In this case, the CE scheme can be applied. Here, X is first compressed into $\hat{X}$, and at the destination, a task-specific estimator then reproduces $\hat{S}$ from $\hat{X}$. The procedure of this scheme is illustrated in Figure 2.

In this paper, quantization is adopted as the compression method, and the MSE distortion is used as the distortion metric. Denoting the state distortion as d_S, we have:

$$d_S(s, \hat{s}) = \|s - \hat{s}\|_2^2. \tag{2}$$

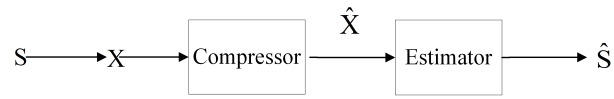


Fig. 2. The procedure of the CE scheme

When designing the quantizer, our objective is to minimize $\mathbb{E}\{d_S(s, \hat{s})\}$. Although S cannot be observed by the quantizer, it turns out that, this indirect optimization objective can be equivalently expressed in a direct form [20] as $\hat{d}_S(x, \hat{s}) = \mathbb{E}\{d_S(S, \hat{s}) \mid X = x\}$.

Denote the quantizer and its input and output by $Q(\cdot)$, $\mathbf{U}$ and $\mathbf{V} = (V_1, V_2, \ldots)$, respectively. In some designs, the quantizer produces $\hat{X}$; in others it directly outputs $\hat{S}$. The rate is $R = H(\mathbf{V})$ when the output components are encoded jointly, and $R = \sum_{i=1}^{K} H(V_i)$ for separate coding, where K is the dimension of $\mathbf{V}$. For analytical convenience, we may employ dithering [21], wherein the dither is denoted by $\mathbf{Z} = (Z, Z, \cdots)$. In this case, $R = H(\mathbf{V}|\mathbf{Z})$ for joint coding, and $R = \sum_{i=1}^{K} H(V_i|Z)$ for separate coding.

III. The Random-Projection-then-Quantization Scheme

In practical applications, quantizers are often universal and task-agnostic, and the QE scheme is used. In some cases, although the specific source model is unknown, it is known that there is a large dimensionality gap between S and X, making the system highly overdetermined. Directly quantizing X then causes significant performance loss. To save bits, the random-projection-then-quantization (RPQ) can be employed, where a K-dimensional ($K \leq n$) random projection of X, denoted by Y, is quantized and used to reconstruct S. Since the relationship between S and X is unknown to the quantizer, a random matrix $\Phi \in \mathbb{R}^{K \times n}$ with i.i.d. Gaussian entries, shared by both the encoder and the decoder, is used to project X into a lower-dimensional space. The procedure of the RPQ scheme is illustrated in Figure 3, establishing the Markov chain: $S \leftrightarrow X \leftrightarrow Y \leftrightarrow \hat{Y} \leftrightarrow \hat{S}$, where $\hat{Y}$ denote the quantization output of Y.

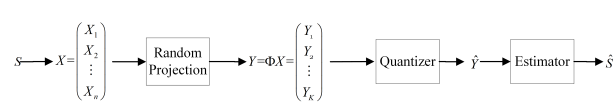


Fig. 3. The procedure of the RPQ scheme

Assume that the estimator receives $\hat{Y} = \hat{y}$. Note that in the quantizer, $\hat{y}$ corresponds to the j-th cell $\mathcal{C}_j$. Hence the output of the estimator $\hat{s}^*$ satisfies:

$$\hat{s}^* = \arg\min_{\hat{s}} \mathbb{E}\{\|\hat{s} - S\|_2^2 | Y \in \mathcal{C}_j\}. \tag{3}$$

For a fixed Φ, S and Y are jointly Gaussian and the relation can be written as $S = BY + N$, where $B = \Sigma_{SY}\Sigma_Y^{-1}$ and N is independent of Y with distribution $\mathcal{N}(0, \Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T)$.

Here Σ_{SY} and Σ_Y denote the cross-covariance matrix between S and Y, and the covariance matrix of Y, respectively. Hence $\mathbb{E}\{\|\hat{S}-S\|_2^2\} = \mathbb{E}\{\|\hat{S}-BY-N\|_2^2\}$. Since the quantization and estimation process is independent of N, $\mathbb{E}\{\|\hat{S}-S\|_2^2\} = \mathbb{E}\{\|\hat{S}-BY\|_2^2\} + \mathbb{E}\{\|N\|_2^2\}$. Denote $L_j(\hat{s}) = \mathbb{E}\{\|\hat{s}-BY\|_2^2 | Y \in \mathcal{C}_j\}$. Hence (3) is equivalent to $\hat{s}^* = \arg\min_{\hat{s}} L_j(\hat{s})$. Then $L_j = \hat{s}^T\hat{s} - 2\mathbb{E}\{Y|Y \in \mathcal{C}_j\}^T B^T \hat{s} + \mathbb{E}\{Y^T B^T BY | Y \in \mathcal{C}_j\}$, and setting $\frac{\partial L_j}{\partial \hat{s}} = 0$ gives

$$\hat{s} = B\mathbb{E}\{Y|Y \in \mathcal{C}_j\}. \tag{4}$$

So when $\forall y \in \mathcal{C}_j$, $Q(y) = \mathbb{E}\{Y|Y \in \mathcal{C}_j\}$, the optimal estimator is $\hat{S} = B\hat{Y}$. Nevertheless, we still adopt this estimator, even though many practical quantizers do not satisfy this condition.

Consider $\mathbb{E}\{\|S-\hat{S}\|_2^2\}$ for a given Φ. From [8], The total distortion can be decomposed into the estimation distortion and the distortion induced by quantization:

$$\mathbb{E}\{\|S-\hat{S}\|_2^2\} = D_0(\Phi) + \mathbb{E}\{\|BY - B\hat{Y}\|_2^2\}, \tag{5}$$

where $D_0(\Phi) = \mathbb{E}\{\|S-BY\|_2^2\}$ denotes the estimation distortion, equal to $\mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T]$.

Theorem 1. *Under high-resolution conditions, using the universal quantization [21], the distortion-rate performance of the RPQ scheme with projection matrix Φ is approximately*

$$D \approx D_0(\Phi) + \frac{\|B\|_F^2}{12} 2^{\frac{2}{K}h(Y)} 2^{-\frac{2}{K}R}, \tag{6}$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $h(\cdot)$ denotes the differential entropy.

When the components of Y are quantized and encoded separately, the distortion-rate performance of the scheme is:

$$D \approx D_0(\Phi) + \frac{\|B\|_F^2}{12} 2^{\frac{2}{K}\sum_{i=1}^{K} h(Y_i)} 2^{-\frac{2}{K}R}. \tag{7}$$

Proof. See Appendix A. □

Remark 1. *Notably, the result $\mathbb{E}\{\|BY - B\hat{Y}\|_2^2\} \approx \|B\|_F^2 \Delta_{QE}/12$ in the aforementioned proof can be generalized to broader scenarios. We apply the estimation function $g(y) = \mathbb{E}\{S|Y=y\}$ to estimate the state S from the low-dimensional representation Y. First, the total distortion can still be decomposed into the estimation distortion and the distortion induced by quantization: $\mathbb{E}\{\|S-\hat{S}\|_2^2\} = \mathbb{E}\{\|S-g(Y)\|_2^2\} + \mathbb{E}\{\|g(Y)-g(\hat{Y})\|_2^2\}$. Under high-resolution condition, $Y - \hat{Y}$ is small. So we can perform a first-order Taylor expansion of $g(Y)$ around $\hat{Y}$, and obtain: $g(Y) \approx g(\hat{Y}) + J_g(\hat{Y})(Y-\hat{Y})$, where $J_g(\hat{Y})$ denotes the Jacobian matrix of g evaluated at $\hat{Y}$. Consequently, we have: $\mathbb{E}\{\|g(Y)-g(\hat{Y})\|_2^2\} \approx \mathbb{E}\{\|J_g(\hat{Y})(Y-\hat{Y})\|_2^2\}$. Specifically in the jointly Gaussian model, $J_g(\hat{Y}) = B$ for every $\hat{Y}$ and the distortion induced by quantization is $\mathbb{E}\{\|BY - B\hat{Y}\|_2^2\}$.*

IV. Case Study

In this section, we consider a specific source model and analyze the rate–distortion performance of the RPQ scheme by taking the expectation over the random projection matrix Φ for a given K. We then study the impact of K on the rate–distortion performance. Throughout this section, the entries of Φ are assumed i.i.d $\mathcal{N}(0, \frac{1}{K})$.

Consider a composite source with $S \sim \mathcal{N}(0,1)$ and $X = hS + W$, where h is an n-dimensional vector and $W \sim \mathcal{N}(\mathbf{0}, \sigma_W^2 I_n)$ is independent of S.

Consider taking expectation over Φ. Using (5), $\mathbb{E}_\Phi\{D\} = \mathbb{E}_\Phi\{D_0(\Phi)\} + \mathbb{E}_\Phi\{\mathbb{E}\{\|BY - B\hat{Y}\|_2^2\}\}$.

Theorem 2. *For the composite source above, projecting X into a K-dimensional vector Y using a Gaussian random matrix Φ yields the expected estimation distortion*

$$\mathbb{E}_\Phi\{D_0(\Phi)\} = \mathbb{E}\left\{\frac{\sigma_W^2}{\sigma_W^2 + \|h\|_2^2 l}\right\}, l \sim \mathrm{Beta}\left(\frac{K}{2}, \frac{n-K}{2}\right),$$

where $\mathrm{Beta}(\cdot,\cdot)$ denotes the Beta distribution.

Proof. See Appendix B. □

Remark 2. *Theorem 2 indicates that increasing K shifts the density of l toward one, making l more concentrated near 1 and reducing the expected estimation distortion.*

A special case arises when $K = n$:

$$\begin{aligned}\mathbb{E}\{D_0(\Phi)\} &= \mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T] \\ &= \mathrm{tr}[\Sigma_S - \Sigma_{SX}\Phi^T(\Phi\Sigma_X\Phi^T)^{-1}\Phi\Sigma_{SX}^T].\end{aligned}$$

Since Φ is invertible with probability one [22],

$$\mathbb{E}\{D_0(\Phi)\} = \mathrm{tr}[\Sigma_S - \Sigma_{SX}\Sigma_X^{-1}\Sigma_{SX}^T] \quad \text{a.s.} \tag{8}$$

Thus, for $K = n$, the expected estimation distortion equals that of directly estimating S from X.

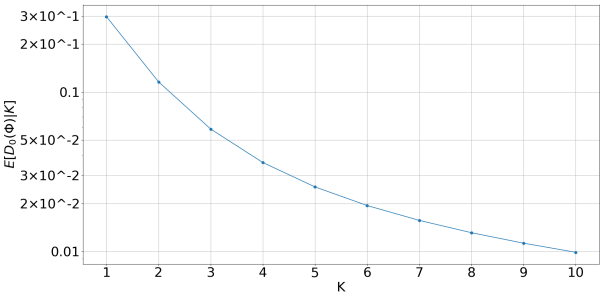
Theorem 1 gives the high-rate distortion–rate performance of the RPQ scheme. When R exceeds a threshold, (6) and (7) are dominated by $D_0(\Phi)$. Intuitively, since Φ is full rank with probability one, a larger K preserves more information about X, reducing $D_0(\Phi)$. Theorem 2 formally establishes this for the composite source considered. Hence, in this regime, dimensionality reduction cannot reduce the total distortion.

Nevertheless, when R does not exceed the threshold, the second term in (5) cannot be neglected. Increasing K lowers the rate per dimension, enlarging scalar quantization intervals and thus increasing quantization distortion. Hence, varying K introduces a trade-off between estimation and quantization distortions. We emphasize that the Gaussian model is used only for closed-form expressions. The trade-off itself does not depend on Gaussianity and is not specific to this model.

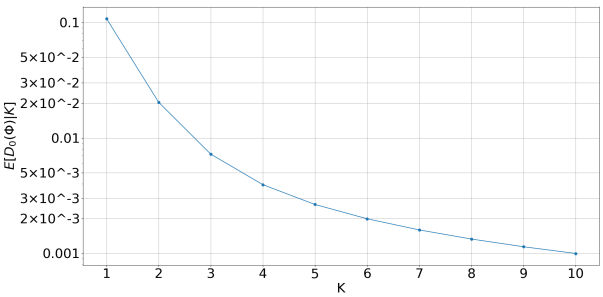
V. Numerical Results

In this section, we evaluate the RPQ scheme for different K and compare it with the QE and EQ schemes. We first conduct numerical simulations on four instances of the source models from Section IV. For the first two sources, we set $n = 10$

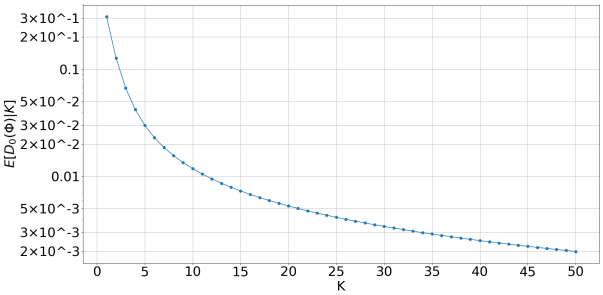
and $\sigma_W^2 = 0.1$ and 0.01. For the other two sources, $n = 50$ with $\sigma_W^2 = 0.1$ and 0.01. For all of the four sources, $h = \mathbf{1}_n$, where $\mathbf{1}_n$ denotes the n-dimensional all-one vector.



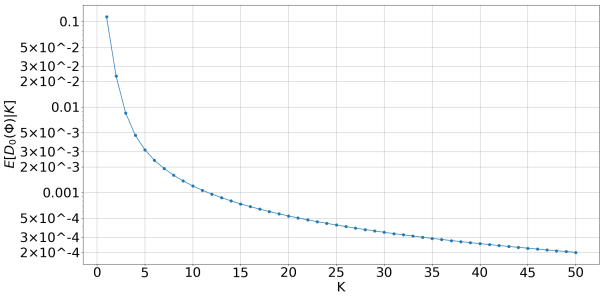
(a): $\mathbb{E}\{D_0(\Phi)\}$ vs K when $n = 10, \sigma_W^2 = 0.1$.



(b): $\mathbb{E}\{D_0(\Phi)\}$ vs K when $n = 10, \sigma_W^2 = 0.01$.



(c): $\mathbb{E}\{D_0(\Phi)\}$ vs K when $n = 50, \sigma_W^2 = 0.1$.



(d): $\mathbb{E}\{D_0(\Phi)\}$ vs K when $n = 50, \sigma_W^2 = 0.01$.

Fig. 4. Relationship between expected estimation distortion $\mathbb{E}\{D_0(\Phi)\}$ and the retained dimension K.

Figure 4 shows the relation between the expected estimation distortion $\mathbb{E}\{D_0(\Phi)\}$ and the retained dimension K for the four source models. The distortion decreases as K increases. When the observation dimension is large or the noise is small, (i.e., high observation redundancy), it drops rapidly. As K becomes large, however, retaining more dimensions provides very marginal additional reduction in estimation distortion. Thus, in many scenarios, random dimensionality reduction can reduce the bit rate while still meeting distortion constraints, unless extremely high precision is required.

When the high-rate condition is not satisfied, the second term in (5) cannot be neglected. Increasing K reduces the bit rate allocated to each dimension, thereby expanding the quantization intervals of the scalar quantizer and consequently increasing quantization distortion. Thus, varying K introduces a trade-off between estimation distortion and quantization distortion.

Figure 5 compares the performance of the RPQ scheme for different K with the QE and EQ schemes, where scalar uniform quantization is used in all schemes. For each K, 10,000 instances of Φ are randomly generated, and for each Φ, 50,000 pairs of (S, X) are simulated. At the quantizer, the number of bits allocated to each dimension is the same.

It can be observed that the EQ scheme outperforms the others, indicating that the knowledge of the semantic relationship between the observation and the state improves the quantizer's rate–distortion performance. However, when the quantizer needs to be task-agnostic, the EQ scheme cannot be implemented. In the low-rate regime, the contribution of quantization distortion becomes more significant. Thus, by employing random projection, although the estimation distortion increases, the total distortion is substantially reduced due to the higher bit rate allocated to each scalar quantizer dimension.

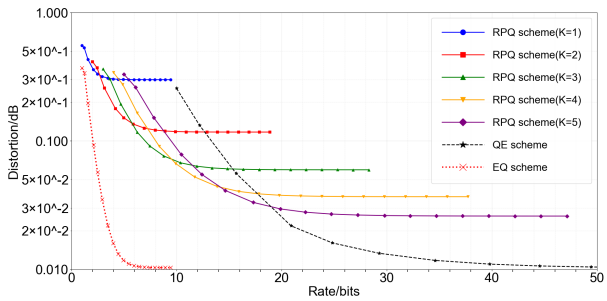
Notably, when n is large or noise is small, RPQ can outperform QE even at higher rates. Intuitively, the intrinsic state is largely concentrated within the high-dimensional observation space. Retaining all n dimensions spreads the bit budget thin, wasting bits on noise or redundancy. By projecting observations onto a K-dimensional subspace, RPQ concentrates bits on a more refined representation. As shown by experimental results, in this regime, quantization coarseness is still the predominant limiting factor.

Another trend observed is that, for the same source, the optimal K increases with the rate, which highlights the trade-off between the two types of distortions.

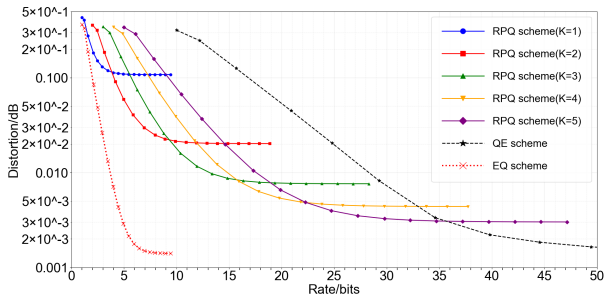
The second simulation is conducted to further test whether a single, task-agnostic random matrix can improve rate-distortion performance across different inference objectives. Consider a composite source with a 2-dimensional state $S = (S_1, S_2)$, where

$$S \sim \mathcal{N}(\mathbf{0}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}), \tag{9}$$

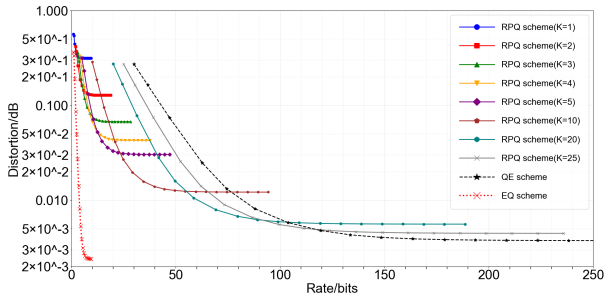
and an observation $X = h_1 S_1 + h_2 S_2 + W$, where h_1 and h_2 are n-dimensional vectors and $W \sim \mathcal{N}(\mathbf{0}, \Sigma_W)$. This gives rise to two inference tasks based on the same observation X. Specifically, the tasks are to reconstruct S_1 and S_2 from the observation, respectively.



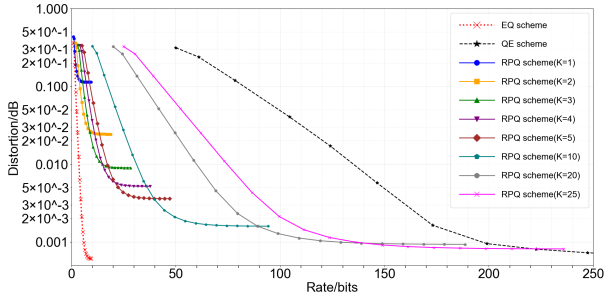
(a): The distortion-rate performances of different schemes when $n = 10, \sigma_W^2 = 0.1$.



(b): The distortion-rate performances of different schemes when $n = 10, \sigma_W^2 = 0.01$.



(c): The distortion-rate performances of different schemes when $n = 50, \sigma_W^2 = 0.1$.



(d): The distortion-rate performances of different schemes when $n = 50, \sigma_W^2 = 0.01$.

Fig. 5. The distortion-rate performances of different schemes.

In our simulation, we set $\rho = 0$, $n = 100$ and $\Sigma_W = 0.01 I_{100}$. The vector h_1 has its first 50 entries equal to 1 and the remaining 50 entries equal to 0, while h_2 has the opposite structure: the first 50 entries are 0 and the last 50 entries are 1. For the RPQ scheme, in each Monte-Carlo trial, a random projection matrix Φ is generated and used by the RPQ encoder to produce quantized measurement $\hat{Y}$. The same projection matrix is used for both tasks within the trial. After obtaining $\hat{Y}$, two separate estimators are applied to produce estimates of S_1 and S_2. This setup allows us to evaluate whether a single task-agnostic RPQ encoder can provide rate savings for multiple downstream inference tasks.

Figure 6 compares the performance of the RPQ scheme with different K with the QE and EQ schemes under the task of reconstructing S_1 and S_2, respectively. The results show that random projection improves the rate–distortion performance for both tasks, even though the projection matrix and the quantization are task-agnostic.

VI. Conclusion

We propose the RPQ scheme to mitigate the rate inefficiency of the conventional QE scheme caused by system overdetermination in certain indirect quantization problems. We analyze its rate-distortion performance, revealing a fundamental trade-off between the estimation distortion and the distortion induced by quantization, which is modulated by the choice of the retained dimension K. Specifically, analytical results characterize the estimation distortion as a function of K, while numerical evaluations demonstrate that the proposed RPQ scheme outperforms the conventional QE scheme in the low-rate regime, and continues to do so at higher rates when the observation dimension is large or the observation noise is small. In future work, we plan to study the relationship between estimation distortion and the dimension K in more general source models. Another promising direction is to investigate how, by introducing a feedback system, both the retained dimension K and the projection matrix can be adaptively adjusted when the exact relationship between the state and observation is unknown.

Appendix A

Denote the components of Y and $\hat{Y}$ as $Y = Y_1, Y_2, \ldots, Y_K$ and $\hat{Y} = \hat{Y}_1, \hat{Y}_2, \ldots, \hat{Y}_K$. So

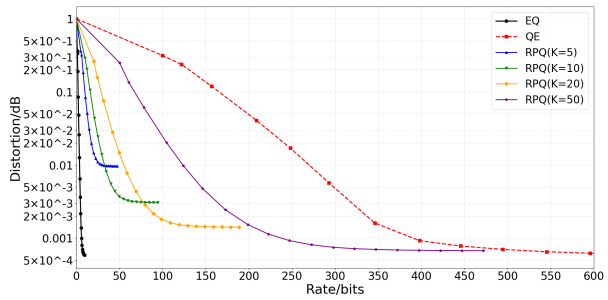
$$\hat{Y} = Q_1(Y + \mathbf{Z}) - \mathbf{Z}, \tag{10}$$

where $Q_1(\cdot)$ represents the uniform scalar quantization, $\mathbf{Z} = (Z, Z, \cdots)$, $Z \sim U(-\sqrt{3\epsilon_{QE}}, \sqrt{3\epsilon_{QE}}]$ and ϵ_{QE} represents the quantization distortion of each dimension, i.e., $\mathbb{E}\{(Q_1(Y_i + Z) - Z - Y_i)^2\} = \epsilon_{QE}$, where $i = 1, 2, \ldots, K$. Hence we have:

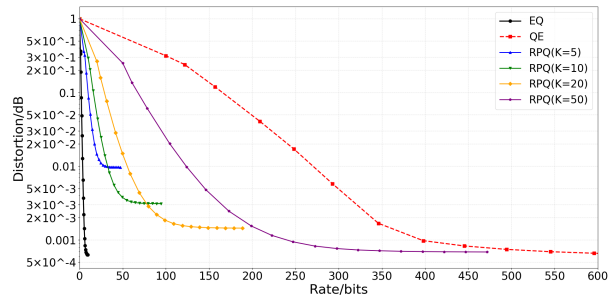
$$\mathbb{E}\{\|BY - B\hat{Y}\|_2^2\} = \mathbb{E}\{\|BQ_1(Y + \mathbf{Z}) - BY - B\mathbf{Z}\|_2^2\}. \tag{11}$$

Denote $Q_1(Y + \mathbf{Z}) = q_1, q_2, \cdots, q_K$. Denote the element in the i-th row and j-th column of B as b_{ij}. Define $d_i = (\sum_{j=1}^{K} b_{ij}(q_j - Y_j - Z))^2$. Therefore,

$$\mathbb{E}\{\|BY - B\hat{Y}\|_2^2\} = \sum_{i=1}^{m} \mathbb{E}\{d_i\}. \tag{12}$$



(a): the distortion–rate performance for reconstructing S_1.



(b): the distortion–rate performance for reconstructing S_2.

Fig. 6. The distortion-rate performances of different schemes for the two tasks.

For each $\mathbb{E}\{d_i\}$,

$$\mathbb{E}\{d_i\} = \mathbb{E}_Y\{\mathbb{E}_{Z|Y}\{d_i \mid Y\}\}. \tag{13}$$

First, we fix Y and solve the expectation w.r.t. Z. Denote this expectation as $\mathbb{E}_Z\{d_i\}$. So

$$\mathbb{E}_Z\{d_i\} = \mathbb{E}_Z\{(\sum_{j=1}^{K} b_{ij}(q_j - Y_j - Z))^2 \mid Y\} \tag{14}$$

For a single squared term, we have

$$\mathbb{E}\{b_{ij}^2(q_j - Y_j - Z)^2 \mid Y\} = b_{ij}^2 \epsilon_{QE}. \tag{15}$$

Consider a single cross term ($j \neq k$), we have

$$\begin{aligned}&\mathbb{E}_Z\{b_{ij}b_{ik}(q_j - Y_j - Z)(q_k - Y_k - Z) \mid Y\}\\ &= b_{ij}b_{ik}\mathbb{E}_Z\{(q_j - Y_j - Z)(q_k - Y_k - Z) \mid Y\}.\end{aligned}$$

Denote $\delta_j = Y_j - q_{min}(Y_j)$ and $q_{min}(Y_j)$ is defined as the greatest quantization value less than or equal to Y_j. Denote $q_{j,min} = q_{min}(Y_j)$. Without loss generality, assume $\delta_j \leq \delta_k$ for simplicity.

$$\begin{aligned}
&\mathbb{E}_Z\{(q_j - Y_j - Z)(q_k - Y_k - Z) \mid Y\}\\
&= \frac{1}{2\sqrt{3\epsilon_{QE}}}\Big(\int_{-\sqrt{3\epsilon_{QE}}}^{\sqrt{3\epsilon_{QE}}-\delta_k} (-z-\delta_j)(-z-\delta_k)dz\\
&\quad + \int_{\sqrt{3\epsilon_{QE}}-\delta_k}^{\sqrt{3\epsilon_{QE}}-\delta_j} (-z-\delta_j)(-z-\delta_k+2\sqrt{3\epsilon_{QE}})dz\\
&\quad + \int_{\sqrt{3\epsilon_{QE}}-\delta_j}^{\sqrt{3\epsilon_{QE}}} (-z-\delta_j+2\sqrt{3\epsilon_{QE}})(-z-\delta_k+2\sqrt{3\epsilon_{QE}})dz\Big)\\
&= \frac{1}{2\sqrt{3\epsilon_{QE}}}\Big(\int_{-\sqrt{3\epsilon_{QE}}}^{\sqrt{3\epsilon_{QE}}-\delta_k} (z+\delta_j)(z+\delta_k)dz\\
&\quad + \int_{\sqrt{3\epsilon_{QE}}-\delta_k}^{\sqrt{3\epsilon_{QE}}-\delta_j} (z+\delta_j)(z+\delta_k)dz\\
&\quad - 2\sqrt{3\epsilon_{QE}}\int_{\sqrt{3\epsilon_{QE}}-\delta_k}^{\sqrt{3\epsilon_{QE}}-\delta_j} z+\delta_j dz\\
&\quad + \int_{\sqrt{3\epsilon_{QE}}-\delta_j}^{\sqrt{3\epsilon_{QE}}} (z+\delta_j-2\sqrt{3\epsilon_{QE}})(z+\delta_k-2\sqrt{3\epsilon_{QE}})dz\Big)\\
&= \frac{1}{2\sqrt{3\epsilon_{QE}}}\Big(\int_{-\sqrt{3\epsilon_{QE}}}^{\sqrt{3\epsilon_{QE}}-\delta_j} (z+\delta_j)(z+\delta_k)dz\\
&\quad + \int_{\sqrt{3\epsilon_{QE}}-\delta_j}^{\sqrt{3\epsilon_{QE}}} (z+\delta_j-2\sqrt{3\epsilon_{QE}})(z+\delta_k-2\sqrt{3\epsilon_{QE}})dz\Big)\\
&\quad - \int_{\sqrt{3\epsilon_{QE}}-\delta_k}^{\sqrt{3\epsilon_{QE}}-\delta_j} (z+\delta_j)dz.
\end{aligned} \tag{16}$$

Denote the two terms of (16) as T_1 and T_2, respectively. Since $\int_{\sqrt{3\epsilon_{QE}}-\delta_j}^{\sqrt{3\epsilon_{QE}}} (z+\delta_j-2\sqrt{3\epsilon_{QE}})(z+\delta_k-2\sqrt{3\epsilon_{QE}})dz = \int_{-\sqrt{3\epsilon_{QE}}-\delta_j}^{-\sqrt{3\epsilon_{QE}}} (z+\delta_j)(z+\delta_k)dz$,

$$\begin{aligned}T_1 &= \frac{1}{2\sqrt{3\epsilon_{QE}}}\int_{-\sqrt{3\epsilon_{QE}}-\delta_j}^{\sqrt{3\epsilon_{QE}}-\delta_j} (z+\delta_j)(z+\delta_k)dz\\ &= \epsilon_{QE}.\end{aligned} \tag{17}$$

$$T_2 = \int_{\sqrt{3\epsilon_{QE}}-\delta_k}^{\sqrt{3\epsilon_{QE}}-\delta_j} (z+\delta_j)dz \\ = (\delta_k-\delta_j)\sqrt{3\epsilon_{QE}} - \frac{1}{2}(\delta_k-\delta_j)^2. \tag{18}$$

Therefore,

$$\mathbb{E}_Z\{(q_j - Y_j - Z)(q_k - Y_k - Z) \mid Y\} \\ = \epsilon_{QE} + \frac{1}{2}(\delta_k-\delta_j)^2 - |\delta_k-\delta_j|\sqrt{3\epsilon_{QE}}. \tag{19}$$

In the high-resolution cases, the conditional distribution within each quantization lattice is approximately uniform. Denote $\delta = \delta_1, \delta_2, \ldots, \delta_K$, $q_{j,min} = q_{min}(Y_j), j = [1:K]$. Let the quantization step size along each dimension be denoted by Δ_{QE}. Then $f(\delta) \approx \frac{1}{\Delta_{QE}^K}$. $\forall j,k \in [1:K]$, we have $f(\delta_j,\delta_k) \approx \frac{1}{\Delta_{QE}^2}$. Hence, we have

$$\mathbb{E}\{(\delta_k-\delta_j)^2\} \approx 2\epsilon_{QE} \tag{20}$$

$$\mathbb{E}\{|\delta_k-\delta_j|\} \approx \frac{2}{3}\sqrt{3\epsilon_{QE}}. \tag{21}$$

Therefore

$$\mathbb{E}\{(q_j - Y_j - Z)(q_k - Y_k - Z) \mid Y\} \approx 0. \tag{22}$$

So

$$\mathbb{E}\{d_i\} = \sum_{j=1}^{K} b_{ij}^2 \epsilon_{QE}, \tag{23}$$

and

$$\mathbb{E}\{\|BY - B\hat{Y}\|_2^2\} = \|B\|_F^2 \epsilon_{QE}. \tag{24}$$

Denote the rate of the scheme with projection matrix Φ and scalar uniform quantization as $R(\Phi, Q_1)$. We have

$$R(\Phi, Q_1) = h(Y) - K\log\Delta_{QE}. \tag{25}$$

Denote the total distortion of the scheme by D. From (5), we have

$$D_0(\Phi) + \|B\|_F^2\epsilon_{QE} = D, \tag{26}$$

so

$$\epsilon_{QE} = \frac{D - D_0(\Phi)}{\|B\|_F^2}. \tag{27}$$

Combined with $\epsilon_{QE} = \frac{\Delta_{QE}^2}{12}$, we have

$$R(\Phi, Q_1) = h(Y) - \frac{K}{2}\log\frac{12(D - D_0(\Phi))}{\|B\|_F^2}. \tag{28}$$

Therefore, the distortion-rate performance of the RPQ scheme under high rate is:

$$D \approx D_0(\Phi) + \frac{\|B\|_F^2}{12} 2^{\frac{2}{K}h(Y)} 2^{-\frac{2}{K}R}. \tag{29}$$

When the K dimensions of Y are quantized and encoded separately, (25) becomes:

$$R(\Phi, Q_1) = \sum_{i=1}^{K}(h(Y_i) - \log\Delta_{QE}) \\ = \sum_{i=1}^{K} h(Y_i) - K\log\Delta_{QE}. \tag{30}$$

Hence the distortion-rate performance of the RPQ scheme in this case is:

$$D \approx D_0(\Phi) + \frac{\|B\|_F^2}{12} 2^{\frac{2}{K}\sum_{i=1}^{K} h(Y_i)} 2^{-\frac{2}{K}R}. \tag{31}$$

Appendix B

Since S and Y are jointly Gaussian, the relationship between the two random variables can be written as $S = BY + Z$. Z is a Gaussian noise independent of Y and follows $\mathcal{N}(0, \Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T)$, where Σ_S, Σ_Y and Σ_{SY} represents the covariance matrices of S, Y and their cross-covariance matrix, respectively. So we have

$$D_0(\Phi) = \mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T]. \tag{32}$$

Specifically in the source described in Section IV, we have $\Sigma_S = 1$, $\Sigma_{SY} = \Sigma_{SX}\Phi^T$ and $\Sigma_Y = \Phi\Sigma_X\Phi^T$. Evidently, we have $\Sigma_{SX} = h^T$ and $\Sigma_X = hh^T + \sigma_W^2 I_n$ in this source. Therefore, $\Sigma_{SY} = h^T\Phi^T$. So we have

$$D_0(\Phi) = 1 - h^T\Phi^T(\Phi\Sigma_X\Phi^T)^{-1}\Phi h. \tag{33}$$

Denote $M = \Phi^T(\Phi\Sigma_X\Phi^T)^{-1}\Phi$. For Σ_Y, we have

$$\Sigma_Y = \Phi hh^T\Phi^T + \sigma_W^2\Phi\Phi^T. \tag{34}$$

Denote $\phi = \Phi h$. So

$$\Sigma_Y = \phi\phi^T + \sigma_W^2\Phi\Phi^T. \tag{35}$$

From Sherman-Morrison formula, we have

$$\Sigma_Y^{-1} = (\sigma_W^2\Phi\Phi^T + \Phi hh^T\Phi^T)^{-1} \\ = \frac{1}{\sigma_W^2}A^{-1} - \frac{A^{-1}\phi\phi^T A^{-1}}{\sigma_W^4 + \sigma_W^2\phi^T A^{-1}\phi}, \tag{36}$$

where $A = \Phi\Phi^T$. So

$$M = \Phi^T\left(\frac{1}{\sigma_W^2}A^{-1} - \frac{A^{-1}\phi\phi^T A^{-1}}{\sigma_W^4 + \sigma_W^2\phi^T A^{-1}\phi}\right)\Phi. \tag{37}$$

Further, we have

$$h^T M h = \frac{1}{\sigma_W^2}\phi^T A^{-1}\phi - \frac{\phi^T A^{-1}\phi\phi^T A^{-1}\phi}{\sigma_W^4 + \sigma_W^2\phi^T A^{-1}\phi} \\ = \frac{\phi^T A^{-1}\phi}{\sigma_W^2 + \phi^T A^{-1}\phi}. \tag{38}$$

Denote $t = \phi^T A^{-1}\phi$, we have

$$\mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T] = 1 - h^T M h \\ = 1 - \frac{t}{\sigma_W^2 + t} \\ = \frac{\sigma_W^2}{\sigma_W^2 + t}. \tag{39}$$

Denote $H = \Phi^T A^{-1}\Phi$, we have $t = h^T H h$. Since H is symmetric and idempotent, it follows that

$$t = h^T H^T H h \\ = \|Hh\|_2^2. \tag{40}$$

For an arbitrary non-zero vector v, consider $\frac{\|Hv\|_2^2}{\|v\|_2^2}$. Without loss of generality, assume that $\|v\|_2^2 = 1$. Denote $e_1 = (1, 0, \ldots, 0)^T$. Then v can be expressed as $v = Ue_1$, where U is an orthogonal matrix. So we have

$$\begin{aligned}\|Hv\|_2^2 &= \|HUe_1\|_2^2 \\ &= e_1^T U^T H U e_1 \\ &= e_1^T (\Phi U)^T ((\Phi U)(\Phi U)^T)^{-1} (\Phi U) e_1.\end{aligned} \tag{41}$$

Denote the row vectors of Φ as $\Phi_1^T, \Phi_2^T, \ldots, \Phi_K^T$, i.e.,

$$\Phi = \begin{pmatrix} \Phi_1^T \\ \Phi_2^T \\ \vdots \\ \Phi_K^T \end{pmatrix}. \tag{42}$$

So

$$\Phi U = \begin{pmatrix} \Phi_1^T U \\ \Phi_2^T U \\ \vdots \\ \Phi_K^T U \end{pmatrix}. \tag{43}$$

For the i-th row vector of Φ, $\Phi_i \sim \mathcal{N}(\mathbf{0}, \frac{1}{K} I_n)$. So

$$\begin{aligned}\mathbb{E}\{U^T \Phi_i \Phi^T U\} &= \frac{1}{K} U^T U \\ &= \frac{1}{K} I_n.\end{aligned} \tag{44}$$

Hence ΦU has the same distribution as Φ. Combined with (41), it follows that $\|He_1\|_2^2$ and $\|Hv\|_2^2$ are identically distributed. Therefore, $\frac{t}{n} = \frac{\|Hh\|_2^2}{\|h\|_2^2}$ and $\frac{\|He_1\|_2^2}{\|e_1\|_2^2} = \|He_1\|_2^2$ are identically distributed. So for simplicity and without loss of generality, we study the distribution of $\|He_1\|_2^2$ in the following derivation.

$$\begin{aligned}\|He_1\|_2^2 &= e_1^T H^T H e_1 \\ &= e_1 \Phi^T (\Phi \Phi^T)^{-1} \Phi e_1.\end{aligned} \tag{45}$$

Define a random matrix $\Psi = \sqrt{K}\Phi$. Then Ψ has i.i.d. entries following $\mathcal{N}(0, 1)$, and $\Psi^T(\Psi\Psi^T)^{-1}\Psi = \Phi^T(\Phi\Phi^T)^{-1}\Phi = H$. So

$$e_1^T H e_1 = e_1^T \Psi^T (\Psi \Psi^T)^{-1} \Psi e_1. \tag{46}$$

Define $a = \Psi e_1$, which is simply the first column of Ψ. Hence

$$e_1^T H e_1 = a^T (\Psi \Psi^T)^{-1} a. \tag{47}$$

Partition Ψ as

$$\Psi = \begin{pmatrix} a & \vdots & W \end{pmatrix}, \tag{48}$$

where the size of W is $K \times (n-1)$. $\Psi\Psi^T = aa^T + WW^T$. Denote $V = WW^T$. From Sherman-Morrison formula, we have

$$(aa^T + V)^{-1} = V^{-1} - \frac{V^{-1} a a^T V^{-1}}{1 + a^T V^{-1} a}. \tag{49}$$

$$\begin{aligned}a^T (aa^T + V)^{-1} a &= a^T V^{-1} a - \frac{(a^T V^{-1} a)^2}{1 + a^T V^{-1} a} \\ &= \frac{a^T V^{-1} a}{1 + a^T V^{-1} a}.\end{aligned} \tag{50}$$

$V \sim W_K(I_K, n-1)$ (W_K denotes the Wishart distribution) and $a \sim \mathcal{N}(\mathbf{0}, I_K)$. So $(n-1)a^T V^{-1} a \sim T^2(K, n-1)$ and $\frac{n-K}{K} a^T V^{-1} a \sim F(K, n-K)$. Since $\frac{n-K}{K} a^T V^{-1} a \sim F(K, n-K)$ follows an $F(K, n-K)$ distribution, it can be expressed as the ratio of two independent chi-square random variables normalized by their respective degrees of freedom, i.e.,

$$\frac{n-K}{K} a^T V^{-1} a = \frac{\frac{U_1}{K}}{\frac{U_2}{n-K}} = \frac{n-K}{K} \frac{U_1}{U_2}, \tag{51}$$

where $U_1 \sim \chi^2(K)$, $U_2 \sim \chi^2(n-K)$, and U_1 and U_2 are independent. After simplification, we obtain:

$$a^T V^{-1} a = \frac{U_1}{U_2}. \tag{52}$$

Hence

$$\|He_1\|_2^2 = \frac{U_1}{U_1 + U_2}. \tag{53}$$

Since $t = \|Hh\|_2^2$ has the same distribution as $\|h\|_2^2 \|He_1\|_2^2$, letting $\|He_1\|_2^2 = l$, with probability density function $f_L(\cdot)$. Denote $s = t/\|h\|_2^2$. Then s has the same distribution as l. Denote the distribution of t and s as $f_T(\cdot)$ and $f_S(\cdot)$, respectively. Thus $f_S(\cdot) = f_L(\cdot)$ and $f_T(t) = \frac{1}{\|h\|_2^2} f_S(s)$. So we have

$$\begin{aligned}\mathbb{E}\{\mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T]\} &= \mathbb{E}\{\frac{\sigma_W^2}{\sigma_W^2 + t}\} \\ &= \int_0^{\|h\|_2^2} f_T(t) \frac{\sigma_W^2}{t + \sigma_W^2} dt \\ &= \int_0^{\|h\|_2^2} \frac{1}{\|h\|_2^2} f_S(s) \frac{\sigma_W^2}{\|h\|_2^2 s + \sigma_W^2} d(\|h\|_2^2 s) \\ &= \int_0^1 f_S(s) \frac{\sigma_W^2}{\|h\|_2^2 s + \sigma_W^2} ds \\ &= \int_0^1 f_L(l) \frac{\sigma_W^2}{\|h\|_2^2 l + \sigma_W^2} dl.\end{aligned} \tag{54}$$

Denote the joint probability density function of U_1 and U_2 by $f_{U_1,U_2}(\cdot)$. We have

$$f_{U_1,U_2}(u_1, u_2) = \frac{1}{2^{\frac{K}{2}}\Gamma(\frac{K}{2})} u_1^{\frac{K}{2}-1} e^{-\frac{u_1}{2}} \frac{1}{2^{\frac{n-K}{2}}\Gamma(\frac{n-K}{2})} u_2^{\frac{n-K}{2}-1} e^{-\frac{u_2}{2}} \tag{55}$$

We have the following transformation:

$$l = \frac{u_1}{u_1 + u_2} \tag{56}$$

and

$$w = u_1 + u_2. \tag{57}$$

The inverse transformation is:

$$u_1 = lw \tag{58}$$

and

$$u_2 = (1-l)w. \tag{59}$$

The Jacobian determinant is

$$J = \begin{vmatrix} w & l \\ -w & 1-l \end{vmatrix} = w. \tag{60}$$

Hence

$$f_{L,W}(l,w) = f_{U_1,U_2}(u_1,u_2)\,|J| = f_{U_1,U_2}(lw,(1-l)w)w. \tag{61}$$

Combined with (55), we have

$$f_{L,W}(l,w) = \frac{1}{\Gamma(\frac{K}{2}))\Gamma(\frac{n-K}{2})2^{\frac{n}{2}}} l^{\frac{K}{2}-1}(1-l)^{\frac{n-K}{2}-1} w^{\frac{n}{2}-1} e^{-\frac{w}{2}}. \tag{62}$$

The marginal pdf of l is obtained by integrating the joint pdf with respect to w:

$$\begin{aligned} f_L(l) &= \int_0^{+\infty} f_{L,W}(l,w)dw \\ &= \frac{l^{\frac{K}{2}-1}(1-l)^{\frac{n-K}{2}-1}}{\Gamma(\frac{K}{2})\Gamma(\frac{n-K}{2})2^{\frac{n}{2}}} \int_0^{+\infty} w^{\frac{n}{2}-1} e^{-\frac{w}{2}} dw \\ &= \frac{l^{\frac{K}{2}-1}(1-l)^{\frac{n-K}{2}-1}}{\Gamma(\frac{K}{2})\Gamma(\frac{n-K}{2})2^{\frac{n}{2}}} 2^{\frac{n}{2}}\Gamma(\frac{n}{2}) \\ &= \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{K}{2})\Gamma(\frac{n-K}{2})} l^{\frac{K}{2}-1}(1-l)^{\frac{n-K}{2}-1}. \end{aligned} \tag{63}$$

Hence

$$\mathbb{E}\{D_0(\Phi) \mid K\} = \mathbb{E}\{\frac{\sigma_W^2}{\sigma_W^2 + \|h\|_2^2 l}\}, l \sim Beta(\frac{K}{2}, \frac{n-K}{2}). \tag{64}$$

References

[1] Y. Bar-Hillel and R. Carnap, "Semantic information," *The British Journal for the Philosophy of Science*, vol. 4, no. 14, pp. 147–157, 1953.

[2] L. Floridi, "Outline of a theory of strongly semantic information," *Minds and Machines*, pp. 197–221, 2004.

[3] D. Gündüz et al., "Beyond transmitting bits: Context, semantics, and task-oriented communications," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 1, pp. 5–41, 2022.

[4] P. Zhang, W. Xu, Y. Liu, X. Qin, K. Niu, S. Cui, et al., "Intellicise wireless networks from semantic communications: A survey, research issues, and challenges," *IEEE Communications Surveys & Tutorials*, vol. 27, no. 3, pp. 2051–2084, 2024.

[5] T. Berger, *Rate Distortion Theory*, Prentice-Hall, Englewood Cliffs, NJ, USA, 1971.

[6] J. Liu, H. V. Poor, I. Song, and W. Zhang, "A rate-distortion analysis for composite sources under subsource-dependent fidelity criteria," *IEEE Journal on Selected Areas in Communications*, 2025.

[7] J. Liu, S. Shao, W. Zhang, and H. V. Poor, "An indirect rate-distortion characterization for semantic sources: General model and the case of gaussian observation," *IEEE Transactions on Communications*, vol. 70, no. 9, pp. 5946–5959, 2022.

[8] J. Wolf and J. Ziv, "Transmission of noisy information to a noisy receiver with minimum distortion," *IEEE Transactions on Information Theory*, vol. 16, no. 4, pp. 406–411, 1970.

[9] R. Dobrushin and B. Tsybakov, "Information transmission with additional noise," *IRE Transactions on Information Theory*, vol. 8, no. 5, pp. 293–304, 1962.

[10] H. Witsenhausen, "Indirect rate distortion problems," *IEEE Transactions on Information Theory*, vol. 26, no. 5, pp. 518–521, 1980.

[11] N. Shlezinger, Y. C. Eldar, and M. R. D. Rodrigues, "Hardware-limited task-based quantization," *IEEE Transactions on Signal Processing*, vol. 67, no. 20, pp. 5223–5238, 2019.

[12] A. Kipnis, S. Rini, and A. J. Goldsmith, "Multiterminal compress-and-estimate source coding," in *2016 IEEE International Symposium on Information Theory (ISIT)*, 2016, pp. 540–544.

[13] A. Kipnis, S. Rini, and A. J. Goldsmith, "The rate-distortion risk in estimation from compressed data," *IEEE Transactions on Information Theory*, vol. 67, no. 5, pp. 2910–2924, 2021.

[14] V. Sulić, J. Perš, M. Kristan, and S. Kovačič, "Dimensionality reduction for distributed vision systems using random projection," in *2010 20th International Conference on Pattern Recognition*, 2010, pp. 380–383.

[15] K. Zhang, L. Zhang, and M. Yang, "Fast compressive tracking," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 10, pp. 2002–2015, 2014.

[16] T. Takiguchi, J. Bilmes, M. Yoshii, and Y. Ariki, "Evaluation of random-projection-based feature combination on speech recognition," in *2010 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2010, pp. 2150–2153.

[17] D.L. Donoho, "Compressed sensing," *IEEE Transactions on Information Theory*, vol. 52, no. 4, pp. 1289–1306, 2006.

[18] P. Li and M. Slawski, "Simple strategies for recovering inner products from coarsely quantized random projections," *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[19] C. Chiu, J. Qin, Y. Zhang, J.i Yu, and Y. Wu, "Self-supervised learning with random-projection quantizer for speech recognition," in *International Conference on Machine Learning*. PMLR, 2022, pp. 3915–3924.

[20] J. Liu, W. Zhang, and H. V. Poor, "A rate-distortion framework for characterizing semantic information," in *2021 IEEE International Symposium on Information Theory (ISIT)*, 2021, pp. 2894–2899.

[21] J. Ziv, "On universal quantization," *IEEE Transactions on Information Theory*, vol. 31, no. 3, pp. 344–347, 1985.

[22] G. W. Anderson, A. Guionnet, and O. Zeitouni, *An introduction to random matrices*, Number 118. Cambridge university press, 2010.